Health Risk Asses
and Causatio

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Chapter Overview

- **Defining Risk**: Understanding risk as the probabilit
occurring, often implying harm.
- **Concepts of Causation**: Reviewing causation from
contemporary perspectives.
- **Quantifying Association**: Using ratio and differenc
quantify associations.

Chapter Overview

- **Assessing Causal Effect**: Applying attributable risk
- **Competing Risks**: Explaining how competing risks patterns.
- **Risk Perception and Communication**: Recognizing risk perception and communication in public healt

1 Risks in Population Heal

- **Central Task:** Assessing health risks core function
- **Broad Usage of " Risk ":** The term describes the pr outcomes, risk factors, and health consequences.
- **Risk Factors vs. Risk Markers:** Distinguishing betw cause disease and those that indicate its presence.
- **Health Risk Appraisal (HRA):** Using quantitative pr mortality and morbidity risks.
- **Health Risk Assessment:** Evaluating scientific data environmental agents ⇔ human exposure.

2 Health Risk Assessment

1. **Hazard Identification**: Identifying harmful
describing toxicity.
2. **Dose-Response Assessment**: Modeling exp
effect size variations.
3. **Exposure Assessment**: Identifying exposed
exposure routes, and dose characteristics.
4. **Risk Characterization**: Integrating data to
likelihood.

② Health Risk Assessment

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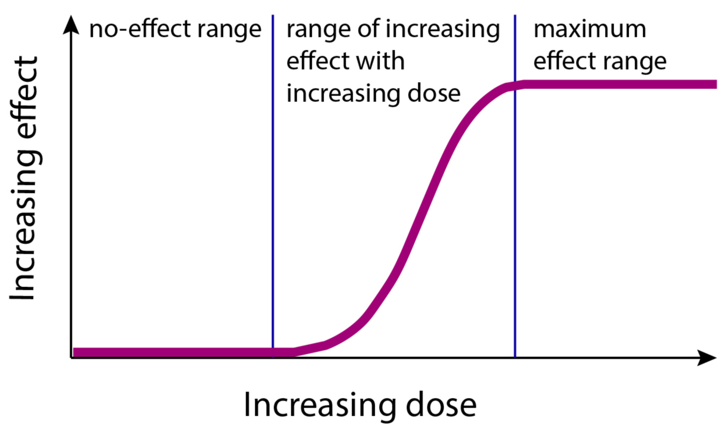
describing toxicity

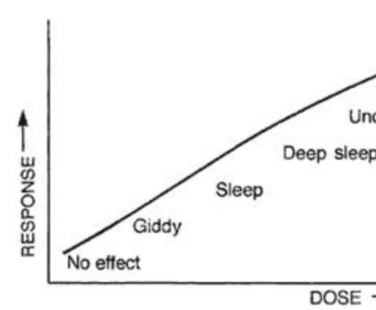
2 Health Risk Assessment

- **Flint resident lives in uncertainty after lead expos**
- **Japan Declares Crisis As Fukushima Reactor Begi**
  And Radiation Levels Soar youtube-link

2 Health Risk Assessment

2. **Dose-Response Assessment**: Modeling exp
effect size variations.





② Health Risk Assessment

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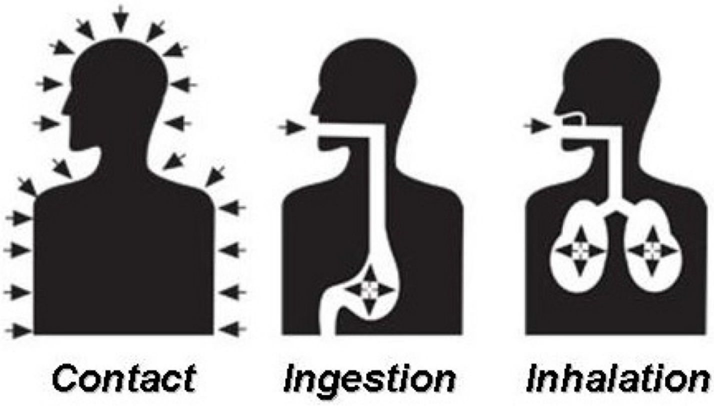

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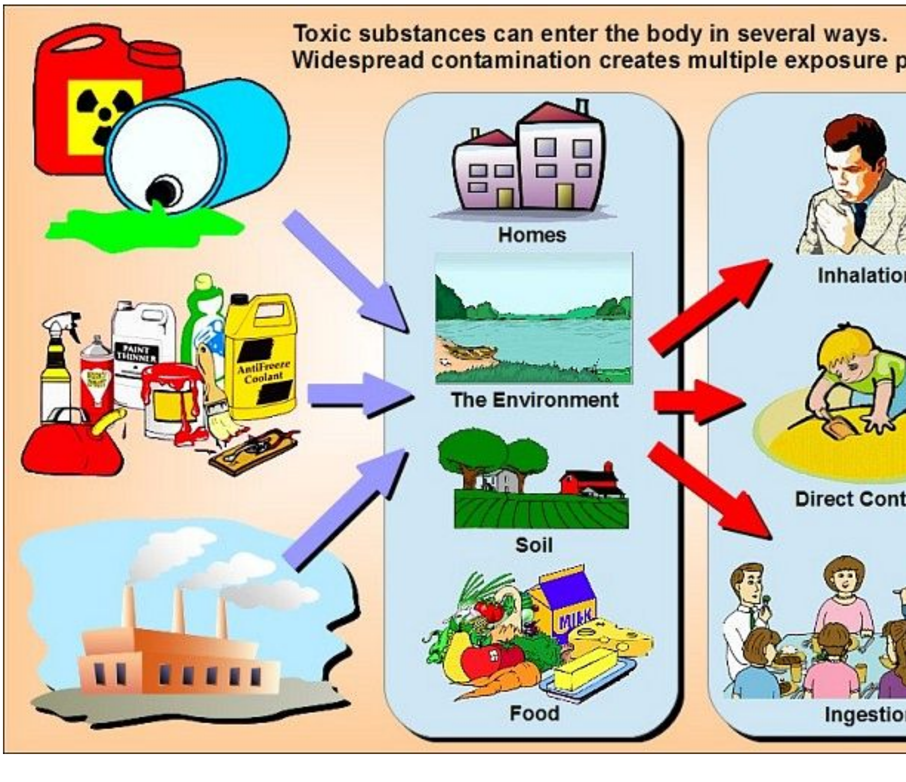

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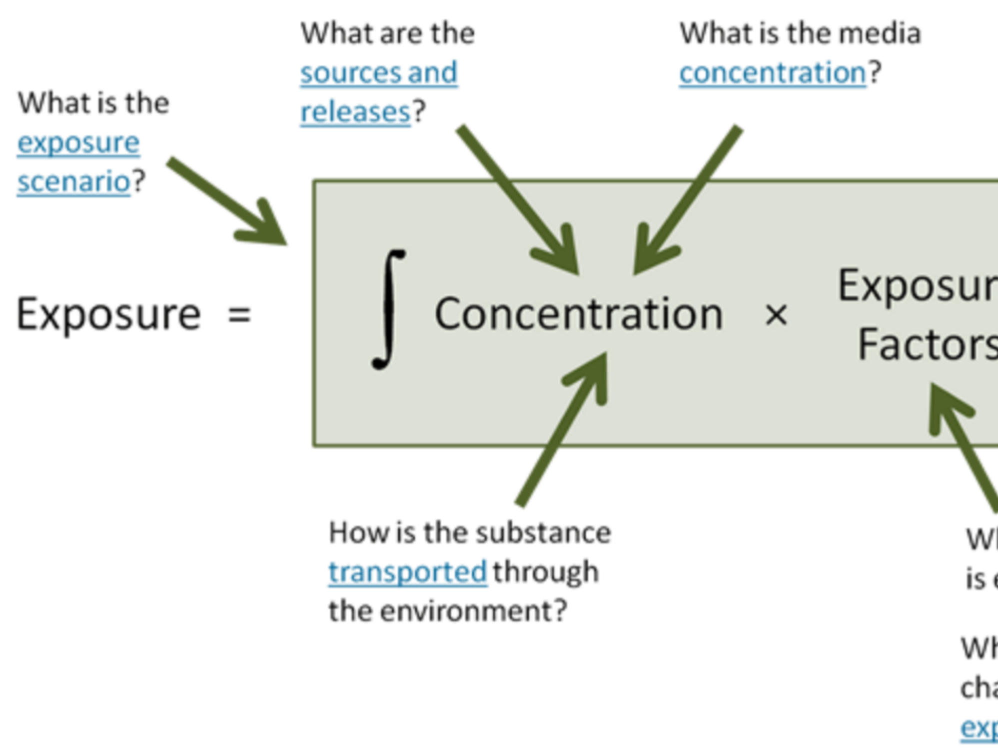

3 Concepts of Causation

- **Necessary Cause**: A cause that must be present for
- **Sufficient Cause**: A cause that inevitably produces
- **Remote vs. Proximate Causes**: Understanding cau proximity.

3 Concepts of Causation

- **Bradford Hill's Criteria:**
  - 1 Strength & Consistency – A strong and repeated association a likelihood of causation.
  - 2 Specificity & Temporality – The cause should lead to a specifi occur **after** the cause.
  - 3 Biological Gradient & Plausibility – A dose-response relations cause should make biological sense.
  - 4 Coherence & Experiment – The evidence should align with exi experiments should support the causal link.
  - 5 Analogy– If similar factors cause similar effects, the associatio

4 Measures of Association

- **Ratio Measures:** Relative risk, risk ratio, rate ratio, o
- **Difference Measures:** Risk difference or attributabl
- **Relative Risk (RR):** $RR = \frac{I_1}{I_0}$
- **Odds Ratio (OR):** $OR = \frac{(a/c)}{(b/d)} = \frac{ad}{bc}$
- **Risk Difference (RD):** $RD = I_1 - I_0$
- **Attributable Risk Among the Exposed [AR(E)]:** $AR(E) = \frac{(I_1 - I_0)}{I_1} = \frac{(RR-1)}{RR}$
- **Population Attributable Risk (PAR):** $PAR = \frac{(I-I}{I}$

5 Competing Risks

- **Definition:** Recognizes that individuals are exposed where one event prevents another.
- **Importance:** Essential for understanding mortality impact.
- **Statistical Approaches:** Methods to estimate death competing risks.

6 Risk Perception and Communication

- **Subjectivity of Risk:** Public perception differs from
- **Social Construction of Risk:** Influences include soc
political factors.
- **Factors Influencing Risk Acceptance:** Fair distribut
alternatives, voluntary vs. imposed risks, and contr
- **Biases in Probability Assessment:** Representativen
availability.
- **Effective Risk Communication:** Public involvemen
honesty, coordination, media engagement, clarity,